AYUSHI RAJ

Python Developer | FastAPI, Browser Automation & API Architecture

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PROFESSIONAL SUMMARY

Python Software Developer and Co-Developer of AstreWork, experienced in FastAPI microservices, headless browser orchestration with Playwright/Puppeteer, and Tree-Sitter AST syntax tree parsing. Implemented subprocess isolation sandboxes to execute untrusted code safely while visualizing code dependency graphs.

TECHNICAL SKILLS

- **Autonomous AI Agents & Intelligent Systems:** Autonomous Browser Agents, Google GenAI SDK, Multi-Model LLM Routing (Groq/Gemini), Dynamic Job-Tailoring Algorithms, ATS Scoring Systems
- **System Design & Full-Stack Architecture:** Next.js 16 (App Router), React 19, Server Components (RSC), Server Actions, RESTful APIs, State Machines (Zustand), Turso LibSQL, Redis 7 Caching
- **Browser Automation & Extensions:** Chrome Extensions (Manifest V3), Shadow DOM Injection, Content Script Orchestration, Playwright, Puppeteer, DOM Parsing, Web Workers
- **Languages & Core Runtimes:** TypeScript, JavaScript, Python, C# (.NET Core), Java (Spring Boot), Node.js, Fastify, SQL
- **UI Engineering & Design Systems:** Tailwind CSS, Vanilla CSS (Glassmorphism & Design Tokens), Monaco Editor, Lucide Icons, Print-Perfect A4 Layouts, Responsive Design
- **Security, Testing & Cloud:** Web Crypto API (AES-GCM, PBKDF2), HMAC-SHA256, JWT, Vitest, Jest, PyTest, Oracle Cloud (OCI), Docker, Git/GitHub

TECHNICAL PROJECTS

Astra Vision - AI Sandbox & Code Parsing Platform (FastAPI, Python, Monaco, Tree-Sitter) Live Demo | GitHub

- **Product Architecture:** Architected an automated code-parsing platform and browser sandbox that allows developers to safely execute untrusted code while visualizing full-stack repository dependency graphs.
- **Sandbox Runtime Challenge:** Built a self-healing Python execution engine with subprocess isolation and security checkpoints to intercept unauthorized OS file system calls and network socket requests in real time.
- **Indexing Performance:** Integrated Tree-Sitter AST compilers and ChromaDB vector search to parse repository syntax nodes, enabling instant code graph dependency analysis under 200ms.

AstreWork — Autonomous Career Platform & AstrePilot Extension (Next.js 16, Chrome MV3, Gemini API) Live Demo | GitHub

- **Co-Developer & Lead Frontend Architect:** Engineered client-side ATS match engine and Gemini-powered scoring studio on Next.js 16, achieving sub-second evaluation across live postings.
- **AstrePilot Chrome Extension (Manifest V3):** Built a cross-browser extension with an isolated glassmorphic Shadow-DOM HUD and HMAC-SHA256 token authorization, executing 1-click ATS application autofill across Greenhouse, Lever, and Ashby in ~15 seconds per application.
- **High-Density Reactive UI Architecture:** Engineered a responsive, accessible workspace in Next.js 16 App Router and React 19 with Zustand state management, real-time ATS audit checklists, and print-perfect A4 CSS layout rendering with zero layout shift.

Alyra Lock - Secure Zero-Knowledge Password Vault (React, TypeScript, Express, MongoDB) Live Demo | GitHub

- **Product Architecture:** Engineered a client-side zero-knowledge password vault ensuring master encryption keys remain entirely isolated inside user browser memory, mitigating cloud database leak risks.
- **Cryptographic Architecture:** Implemented Web Crypto API primitives utilizing AES-GCM (256-bit) payload encryption and PBKDF2 key derivation over 100,000 hashing iterations for key security.
- **Performance & Security:** Developed high-throughput Express.js REST API sync endpoints with JWT authentication and strict CORS headers, keeping vault database sync latency under 200ms.

TECHNICAL SIMULATIONS & VIRTUAL EXPERIENCES

Commonwealth Bank (CommBank) - Software Development Virtual Simulation June 2026

- **Problem & Scope:** Diagnosed and resolved silent data overwrite bugs across high-traffic C#/.NET Core financial API controllers handling MongoDB document updates across financial services endpoints.
- **Action Taken:** Extended C# Web API controllers with $set atomic operators for partial payload updates and modernized an interactive React/Redux Goal Manager UI component for banking clients.
- **Key Outcome:** Authored comprehensive automated test suites using xUnit and Moq, covering complex boundary conditions to guarantee high reliability and transactional consistency.

Y Combinator (YC Startup - Shiptivity) - Software Development Virtual Simulation June 2026

- **Problem & Scope:** Fixed key constraint collisions and slow re-rendering performance on a real-time drag-and-drop Kanban workflow board used by engineering teams to track task state transitions.
- **Action Taken:** Architected dynamic SQLite priority reordering logic using atomic transactions to update priority ranks sequentially and integrated Dragula drag-and-drop state hooks in React.
- **Key Outcome:** Reduced UI component re-render cycles by 30% and patched OpenSSL compilation bottlenecks in legacy Node.js Webpack configurations to optimize runtime builds.

Walmart USA - Advanced Software Development Virtual Simulation May 2026

- **Problem & Scope:** Solved indexing overhead and performance bottlenecks in high-volume retail inventory processing queues handling unnormalized data ingestion from legacy spreadsheets.
- **Action Taken:** Implemented a generic K-ary Max Heap structure in Java using fast bitwise shift operators (<<, >>>) and built 3NF database schemas with design pattern abstractions.
- **Key Outcome:** Accelerated queue indexing calculations by 35% and engineered automated Python ETL pipelines using csv and sqlite3 to clean and consolidate disparate spreadsheets.

EDUCATION

Bachelor of Computer Applications (BCA) - Amity University Noida Expected July 2028

Relevant Coursework: Data Structures & Algorithms (DSA) • Database Management Systems (DBMS) • Operating Systems (OS) • Computer Networks • Object-Oriented Programming (OOP)